

Core Programming

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494	Identifying ‘Soft Trends’ – Topics 7
501	Technology Directions – Topics 7
503	Tools-Related Trends – Topics 2

	Summary
	Concluding Comments
503	The Importance of Software – Revisited
504	People and The Way They Build Systems
505	New Modes for Representing Information
506	The Long View
507	The Software Engineer’s Responsibility
508	A Final Comment
	Part-6: Appendix
509	An Introduction to UML
510	Object Oriented Concepts

Pure Mathematics – 1, 2, and 3

	Pure Mathematics (A-Level)
	P1: Algebra
01	Background Algebra
02	Linear Equations
03	Changing the Subject of a Formula
04	Quadratic Equations
05	Solving Quadratic Equations
06	Equations that cannot be Factorized
07	The Graphs of Quadratic Function
08	The Quadratic Formula
09	Inequalities
	P2: Algebra
10	Operations with Polynomials
11	Solution of Polynomial Equations
12	The Modulus Function
	P3: Further Algebra
13	The General Binomial Expansion
14	Review of Algebraic Functions
15	Partial Functions
16	Using Partial Functions with The Binomial Expansion
	P1: Co-Ordinate Geometry
17	Co-Ordinates
18	Plotting, Sketching and Drawing
19	The Gradient of a Line
20	The Distance Between Two Points
21	The Mid-Point of a Line Joining Two Points
22	The Equation of a Straight Line
23	Finding the Equation of a Line
24	The Intersection of Two Line
25	Drawing Curves
26	The Intersection of A Line and A Curve
	P1: Sequences and Series
27	Definitions and Notation
28	Arithmetic Progressions
29	Geometric Progressions
30	Binomial Expansions
	P1: Functions
31	The Language of Functions
32	Composite Functions
33	Inverse Functions
	P1: Differentiation
34	The gradient of a Curve
35	Finding the Gradient of a Curve
36	Finding the Gradient from First Principles
37	Differentiating by Using Standard Results
38	Using Differentiation
39	Tangents and Normals
40	Maximum and Minimum Points
41	Increasing and Decreasing Functions
42	Points of Inflection
43	The Second Derivative
44	Applications

45	The Chain Rule
	P2: Differentiation
46	The Product Rule
47	The Quotient Rule
48	Differentiating Natural Logarithms and Exponentials
49	Differentiating Trigonometrical Functions
50	Differentiating Functions Defined Implicitly
51	Parametric Equations
52	Parametric Differentiation
	P3: Differential Equations
53	Forming Differential Equations from Rates of Change
54	Solving Differential Equations
	P1: Integration
55	Reversing Differentiation
56	Finding the Area Under A Curve
57	Area as The Limit of A Sum
58	Areas Below the X Axis
59	The Area Between Two Curves
60	The Area Between A Curve and the Y Axis
61	The Reverse Chain Rule
62	Improper Integrals
63	Finding Volumes by Integration
	P2: Integration
64	Integrals Involving The Exponential Function
65	Integrals Involving The Natural Logarithm Function
66	Integrals Involving Trigonometrical Functions
67	Numerical Integration
	P3: Further Integration
68	Integration by Substitution
69	Integrals Involving Exponentials and Natural Logarithms
70	Integrals Involving Trigonometrical Function
71	The Use of Partial Fractions in Integration
72	Integration by Parts
73	General Integration
	P1: Trigonometry
74	Trigonometry Background
75	Trigonometrical Functions
76	Trigonometrical Functions for Angles of any Size
77	The sine and cosine Graphs
78	The tangent graph
79	Solving Equations using Graphs of Trigonometrical Functions
80	Circular Measure
81	The Length of An ARC of A Circle
82	The Area of A Sector of A Circle
83	Other Trigonometrical Functions
	P2: Trigonometry
84	Reciprocal Trigonometrical Functions
85	Compound-Angle Formulae
86	Double-Angle Formulae
87	The Forms $r\cos$, $r\sin$
88	The General Solutions of Trigonometrical Equations
	P1: Vectors
89	Vectors in Two Dimensions
90	Vectors in Three Dimensions
91	Vectors Calculations

92	The Angle Between Two Vectors
	P3: Vectors
93	The Vector Equation of a Line
94	The Intersection of Two Lines
95	The Angle Between Two Lines
96	The Perpendicular Distance from A Point to a Line
97	The Vector Equation of a Plane
98	The Intersection of A Line and A Plane
99	The Distance of A Point from A Plane
100	The Angle Between A Line and A Plane
101	The Intersection of Two Planes
	P2: Logarithms and Exponentials
102	Logarithms
103	Exponential Functions
104	Modelling Curves
105	The Natural Logarithm Functions
106	The Exponential Function
	P2: Numerical Solution of Equations
107	Interval Estimation – Change-of-Sign Methods
108	Fixed-Point Iteration
	P3: Complex Numbers
109	The Growth of the Number System
110	Working with Complex Numbers
111	Representing Complex Numbers Geometrically
112	Sets of Points in An Argand Diagram
113	The Modulus-Argument form of Complex Numbers
114	Sets of Points Using The Polar Form
115	Working with Complex Numbers in Polar Form
116	Complex Exponents
117	Complex Numbers and Equations

Discrete Mathematics

Discrete Mathematics: Reference-1

	Discrete Mathematics – Schaum's Outlines
	Set Theory
01	Introduction
02	Sets and Elements, Subsets
03	Ven Diagrams
04	Algebra of Sets, Duality
05	Finite Sets, Counting Principle
06	Classes of Sets, Power Sets, Partitions
07	Mathematical Introduction
	Solved Problems
	Supplementary Problems
	Relations
08	Introduction
09	Product Sets
10	Relations
11	Pictorial Representatives of Relations
12	Composition of Relations
13	Types of Relations
14	Closure Properties
15	Equivalence Relations
16	Partial Ordering Relations
	Solved Problems
	Supplementary Problems
	Functions and Algorithms
17	Introduction
18	Functions
19	One-to-One, Onto, and Invertible Functions
20	Mathematical Functions, Exponential and Logarithmic Functions
21	Sequences, Indexed Classes of Sets
22	Recursively Defined Functions
23	Cardinality
24	Algorithms and Functions
25	Complexity of Algorithms
	Solved Problems
	Supplementary Problems
	Logic and Propositional Calculus
26	Introduction
27	Propositions and Compound Statements
28	Basic Logical Operations
29	Propositions and truth Tables
30	Tautologies and Contradictions
31	Logical Equivalence
32	Algebra of Propositions
33	Conditional and Bi-conditional Statements
34	Arguments
35	Propositional Functions, Quantifiers
36	Negation of Quantified Statements
	Solved Problems
	Supplementary Problems
	Techniques of Counting

37	Introduction
38	Basic Counting Principles
39	Mathematical Functions
40	Permutations
41	Combinations
42	The Pigeonhole Principle
43	The Inclusion-Exclusion Principle
44	Tree Diagrams
	Solved Problems
	Supplementary Problems
	Advanced Counting Techniques, Recursion
45	Introduction
46	Combinations with Repetitions
47	Ordered and Unordered Partitions
48	Inclusion – Exclusion Principle Revisited
49	Pigeonhole Principle Revisited
50	Recurrence Relations
51	Linear Recurrence Relations with Constant Coefficients
52	Solving Second-Order Homogeneous Linear Recurrence Relations
53	Solving General Homogeneous Linear Recurrence Relations
	Solved Problems
	Supplementary Problems
	Probability
54	Introduction
55	Sample Space and Events
56	Finite Probability Spaces
57	Conditional Probability
58	Independent Events
59	Independent Repeated Trials, Binomial Distribution
60	Random Variables
61	Chebyshev's Inequality, Law of Large Numbers
	Solved Problems
	Supplementary Problems
	Graph Theory
62	Introduction, Data Structures
63	Graphs and Multigraphs
64	Subgraphs, Isomorphic and Homeomorphic Graphs
65	Paths, Connectivity
66	Traversable and Eulerian Graphs, Bridges of Königsberg
67	Labeled and Weighted Graphs
68	Complete, Regular, and Bipartite Graphs
69	Tree Graphs
70	Planar Graphs
71	Graph Colorings
72	Representing Graphs in Computer Memory
73	Graph Algorithms
74	Traveling-Salesman Problem
	Solved Problems
	Supplementary Problems
	Directed Graphs
75	Introduction
76	Directed Graphs
77	Basic Definitions
78	Rooted Trees
79	Sequential Representation of Directed Graphs

80	Warshall's Algorithm, Shortest Paths
81	Linked Representation of Directed Graphs
82	Graph Algorithms: Depth-First and Breadth-First Searches
83	Directed Cycle-Free Graphs, Topological Sort
84	Pruning Algorithm for Shortest Path
	Solved Problems
	Supplementary Problems
	Binary Trees
85	Introduction
86	Binary Trees
87	Complete and Extended Binary Trees
88	Representing Binary Trees in Memory
89	Traversing Binary Trees
90	Binary Search Trees
91	Priority Queues, Heaps
92	Path Lengths, Huffman's Algorithm
93	General (Ordered Rooted) Trees Revisited
	Solved Problems
	Supplementary Problems
	Properties of the Integers
94	Introduction
95	Order and Inequalities, Absolute Value
96	Mathematical Induction
97	Division Algorithm
98	Divisibility, Primes
99	Greatest Common Divisor, Euclidean Algorithm
100	Fundamental theorem of Arithmetic
101	Congruence Relation
102	Congruence Equations
	Solved Problems
	Supplementary Problems
	Languages, Automata, Grammars
103	Introduction
104	Alphabet, Words, Free Semigroup
105	Languages
106	Regular Expressions, Regular Languages
107	Finite State Automata
108	Grammars
	Solved Problems
	Supplementary Problems
	Finite State Machines and Turning Machines
109	Introduction
110	Finite State Machines
111	Godel Numbers
112	Turning Machines
113	Computable Functions
	Solved Problems
	Supplementary Problems
	Ordered Sets and Lattices
114	Introduction
115	Ordered Sets
116	Hasse Diagrams of Partially Ordered Sets
117	Consistent Enumeration
118	Supremum and Infimum
119	Isomorphic (Similar) Ordered Sets

120	Well-Ordered Sets
121	Lattices
122	Bounded Lattices
123	Distributive Lattices
124	Complements, Complemented Lattices
	Solved Problems
	Supplementary Problems
	Boolean Algebra
125	Introduction
126	Basic Definitions
127	Duality
128	Basic Theorems
129	Boolean Algebras as Lattices
130	Representation Theorem
131	Sum-of-Products Form for Sets
132	Sum-of-Products Form for Boolean Algebras
133	Minimal Boolean Expressions, Prime Implicants
134	Logic Gates and Circuits
135	Truth Tables, Boolean Functions
136	Karnaugh Maps
	Solved Problems
	Supplementary Problems
	Appendix: Vectors and Matrices
137	Introduction
138	Vectors
139	Matrices
140	Matrix Addition and Scalar Multiplication
141	Matrix Multiplication
142	Transpose
143	Square Matrices
144	Invertible (Nonsingular) Matrices, Inverses
145	Determinants
146	Elementary Row Operations, Gaussian Elimination (Optional)
147	Boolean (Zero-One) Matrices
	Solved Problems
	Supplementary Problems
	Appendix: Algebraic Systems
148	Introduction
149	Operations
150	Semigroups
151	Groups
152	Subgroups, Normal Subgroups, and Homomorphisms
153	Rings, Integral Domains, and Fields
154	Polynomials Over a Field
	Solved Problems
	Supplementary Problems

Discrete Mathematics: Reference-2

	Discrete Mathematics and Its Applications
	The Foundations: Logic and Proofs
06	Propositional Logic – Topics 6
12	Applications of Propositional Logic – Topics 6
19	Propositional Equivalences – Topics 7
32	Predicates and Quantifiers – Topics 13
39	Nested Quantifiers – Topics 7
47	Rules of Inference – Topics 8
56	Introduction to Proofs – Topics 9
66	Proof Methods and Strategy – Topics 10
	End-of-Chapter Material
	Basic Structures: Sets, Functions, Sequences, Sums
74	Sets – Topics 8
79	Set Operations – Topics 5
85	Functions – Topics 6
90	Sequences and Summations – Topics 5
93	Cardinality of Sets – Topics 3
97	Matrices – Topics 4
	End-of-Chapter Material
	Algorithms
103	Algorithms – Topics 6
108	The Growth of Functions – Topics 5
113	Complexity of Algorithms – Topics 5
	End-of-Chapter Material
	Number Theory and Cryptography
118	Divisibility and Modular Arithmetic – Topics 5
122	Integer Representations and Algorithms – Topics 4
130	Primes and Greatest Common Divisors – Topics 8
137	Solving Congruencies – Topics 7
140	Applications of Congruencies – Topics 3
149	Cryptography – Topics 9
	End-of-Chapter Material
	Induction and Recursion
157	Mathematical Induction – Topics 8
162	Strong Induction and Well-Ordering – Topics 5
167	Recursive Definitions and Structural Induction – Topics 5
171	Recursive Algorithms – Topics 4
176	Program Correctness – Topics 5
	End-of-Chapter Material
	Counting
182	The Basics of Counting – Topics 6
185	The Pigeonhole Principle – Topics 3
188	Permutations and Combinations – Topics 3
191	Binominal Coefficients and Identities – Topics 3
196	Generalized Permutations and Combinations – Topics 5
199	Generating Permutations and Combinations – Topics 3
	End-of-Chapter Material
	Discrete Probability
203	An Introduction to Discrete Probability – Topics 4
213	Probability Theory – Topics 10
216	Bayes' Theorem – Topics 3
224	Expected Value and Variance – Topics 8

	End-of-Chapter Material
	Advanced Counting Techniques
227	Applications of Recurrence Relations – Topics 3
230	Solving Linear Recurrence Relations – Topics 3
232	Divide-and-Conquer Algorithms and Recurrence Relations – Topics 2
237	Generating Functions – Topics 5
239	Inclusion-Exclusion – Topics 2
244	Applications of Inclusion-Exclusion – Topics 5
	End-of-Chapter Material
	Relations
249	Relations and Their Properties – Topics 5
255	n-ary Relations and Their Applications – Topics 6
258	Representing Relations – Topics 3
263	Closures of Relations – Topics 5
267	Equivalence Relations – Topics 4
273	Partial Orderings – Topics 6
	End-of-Chapter Material
	Graphs
274	Graphs and Graph Models – Topics 1
281	Graph Terminology and Special Types of Graphs – Topics 7
287	Representing Graphs and Graph Isomorphism – Topics 6
294	Connectivity – Topics 7
298	Euler and Hamilton Paths – Topics 4
301	Shortest-Path Problems – Topics 3
303	Planar Graphs – Topics 3
305	Graph Coloring – Topics 2
	End-of-Chapter Material
	Trees
308	Introduction to Trees – Topics 3
313	Applications of Trees – Topics 5
317	Tree Traversal – Topics 4
322	Spanning Trees – Topics 5
324	Minimum Spanning Trees – Topics 2
	End-of-Chapter Material
	Boolean Algebra
329	Boolean Functions – Topics 5
331	Representing Boolean Functions – Topics 2
335	Logic Gates – Topics 4
339	Minimization of Circuits – Topics 4
	End-of-Chapter Material
	Modeling Computation
344	Languages and Grammars – Topics 5
346	Finite-State Machines with Output – Topics 2
351	Finite-State Machines with No Output – Topics 5
356	Language Recognition – Topics 5
363	Turing Machines – Topics 7
	End-of-Chapter Material
	Appendices
	Axioms for the Real Numbers and the Positive Integers
	Exponential and Logarithmic Functions
	Pseudocode